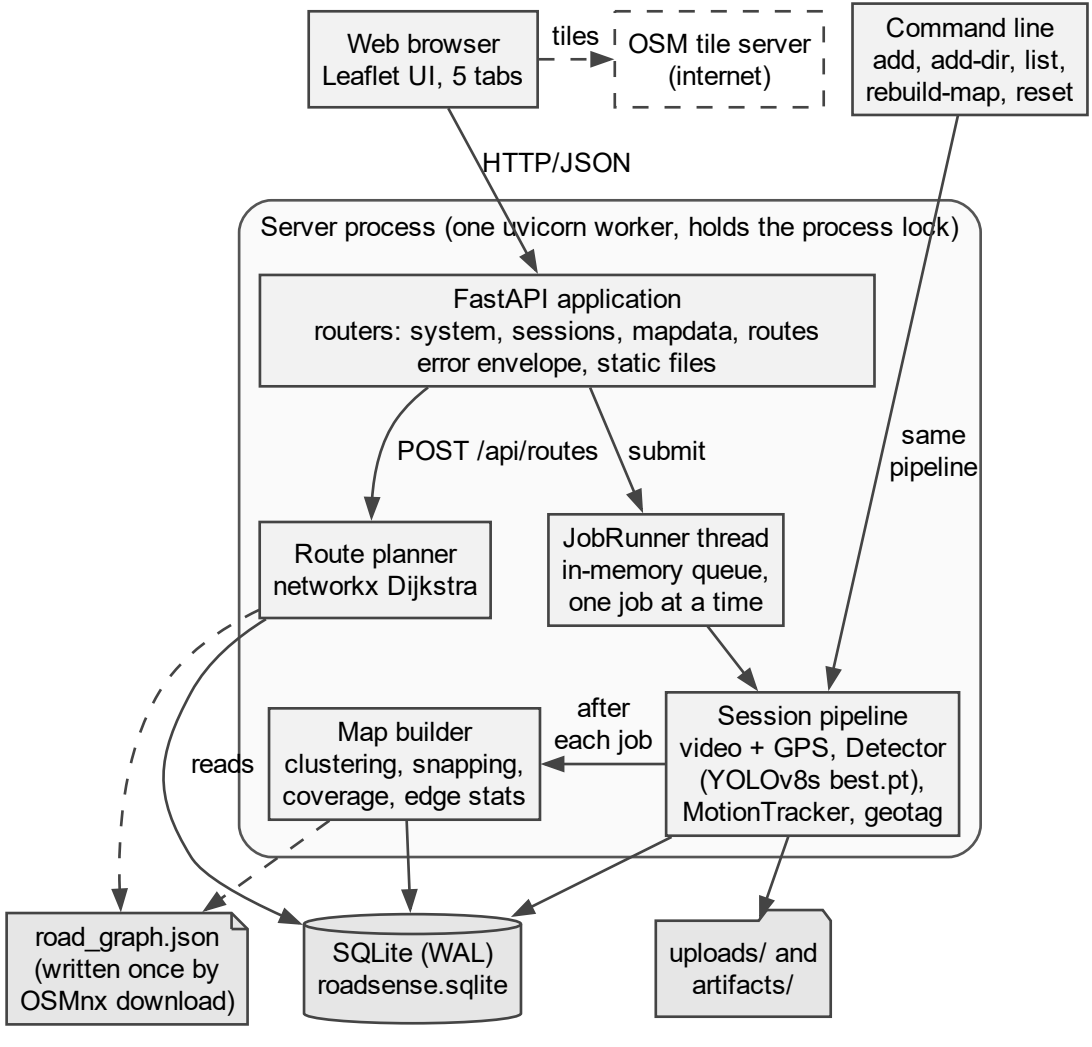
Web browser
Leaflet UI, 5 tabs
tiles
OSM tile server
(internet)
Command line
add, add-dir, list,
rebuild-map, reset
HTTP/JSON
Server process (one uvicorn worker, holds the process lock)
FastAPI application
routers: system, sessions, mapdata, routes
error envelope, static files
POST /api/routes
submit
same
pipeline
Route planner
networkx Dijkstra
JobRunner thread
in-memory queue,
one job at a time
after
each job
Map builder
clustering, snapping,
coverage, edge stats
Session pipeline
video + GPS, Detector
(YOLOv8s best.pt),
MotionTracker, geotag
reads
road_graph.json
(written once by
OSMnx download)
SQLite (WAL)
roadsense.sqlite
uploads/ and
artifacts/